BIST Algorithmic Trading Challenge

Python Backtesting Assignment

Objective

Design and implement one algorithmic trading strategy in Python for the following six BIST stocks: AKBNK, ASELS, TUPRS, TCELL, FROTO, and EREGL.

- Backtest period: 1 January 2025 - 1 October 2026
- Initial capital: 100,000 TL per stock
- The same core strategy must be applied to all six stocks.
- Goal: outperform the benchmark result for every stock individually.

Benchmark Results

Stock	Initial Capital	Benchmark Net Profit	Benchmark Final Capital
AKBNK	100,000 TL	+84,000 TL	184,000 TL
ASELS	100,000 TL	+239,000 TL	339,000 TL
TUPRS	100,000 TL	+72,000 TL	172,000 TL
TCELL	100,000 TL	-18,000 TL	82,000 TL
FROTO	100,000 TL	+3,000 TL	103,000 TL
EREGL	100,000 TL	+39,000 TL	139,000 TL

Important: overall portfolio profit is not used for passing the challenge. Each stock must beat its own benchmark individually.

Example:
Suppose your strategy produces:

- AKBNK: **+90,000 TL** → PASS, because the benchmark is +84,000 TL
- ASELS: **+250,000 TL** → PASS, because the benchmark is +239,000 TL
- TUPRS: **+60,000 TL** → FAIL, because the benchmark is +72,000 TL

Even if your total portfolio profit is higher than the benchmark total, the challenge is **not passed** because TUPRS did not beat its own benchmark.

You must beat the benchmark for all six stocks individually.

Student Task

- Create one algorithmic trading system and apply it to all six stocks. Create one algorithmic trading system and apply it to all six stocks. You may either use one of the four strategies covered in class or design and implement your own trading system.
- You may use indicators such as moving averages, RSI, Bollinger Bands, momentum, volume, ATR, support/resistance, breakouts, mean reversion, trailing stops, or machine-learning features.
- Trade decisions must be generated by Python rules, not by manually selected dates.
- At least 3 completed trades must be generated for every stock.
- You must design and implement a trading strategy and clearly present the complete results. Your submission must include **your source code, all buy and sell times, the exact buy and sell prices for each trade, the profit or loss of each completed trade, and a chart showing the trades and their resulting profits/losses.**

Required Python Workflow

- Load historical price data.
- Calculate indicators or features.
- Generate BUY and SELL signals.
- Apply position sizing and risk management.
- Run the backtest.
- Calculate performance metrics.
- Compare each stock with the benchmark.

- You may also use and improve any of the four strategies covered in class:
  - **Volume Bubbles Strategy**
  - **Liquidity Swings Strategy**
  - **Support & Resistance Channels Strategy**
  - **Dual-Phase Reversal Strategy**
- You are allowed to modify, extend, or improve these strategies before applying them to the six stocks.

No Look-Ahead Rule

The strategy must use only information available at the time of the trade. Future prices, future returns, or manually selected best entry/exit dates are not allowed.

Risk Management Requirement

The strategy must include at least one risk-management mechanism, such as a stop-loss, ATR stop, trailing stop, Chandelier Exit, maximum holding period, or risk-based position sizing.

Required Metrics

- Net Profit
- Final Capital
- Total Trades
- Winning Trades
- Losing Trades
- Win Rate
- Maximum Drawdown
- Profit Factor
- Average Trade

Final Analysis Question

Explain why the same strategy performed differently across the six stocks.

Discuss factors such as trending vs. sideways markets, volatility, false breakouts, momentum strength, reversal behavior, volume characteristics, and stop-loss frequency.

Benchmark Comparison

Requirement: the strategy must outperform the benchmark result for each of the six stocks individually. A higher total portfolio profit does not compensate for failing to beat the benchmark on any single stock. A stock passes only when its final capital is higher than that stock's benchmark final capital shown above, while respecting all assignment rules.

You must submit your **Python Jupyter Notebook with all code cells and console/output cells clearly visible**. Your notebook must also include **figures that visually show your buy and sell orders, entry and exit prices, and the profit or loss for each trade**. For AKBNK, the result according to the **Volume Bubbles Strategy** is shown below

